1. Write a code snippet to turn Bluetooth ON using BluetoothAdapter.
2. How can you get a list of paired Bluetooth devices programmatically?

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

     <uses-permission
android:name="android.permission.BLUETOOTH" />
     <uses-permission
android:name="android.permission.BLUETOOTH_ADMIN" />
     <uses-permission
android:name="android.permission.BLUETOOTH_CONNECT" />

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.AppPr20Bluetooth">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category
android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

XML CODE:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/main"
```

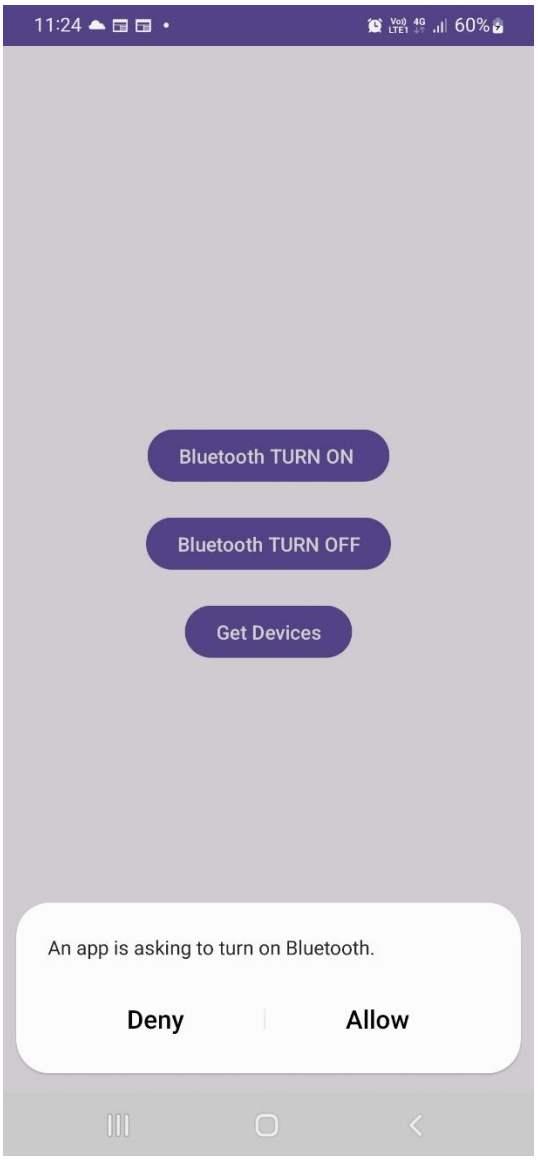
```
    android:orientation="vertical"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="20dp"
        android:text="Bluetooth TURN ON "
  />

    <Button
        android:id="@+id/button2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="20dp"
        android:text="Bluetooth TURN OFF" />

    <Button
        android:id="@+id/button3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="20dp"
        android:text="Get Devices" />

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="" />

</LinearLayout>
```

JAVA CODE:

```
package com.example.apppr20bluetooth;

import android.bluetooth.BluetoothAdapter;
import android.bluetooth.BluetoothDevice;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;

import androidx.activity.result.ActivityResultLauncher;
import androidx.activity.result.contract.ActivityResultContract;
import androidx.activity.result.contract.ActivityResultContracts;
import androidx.appcompat.app.AppCompatActivity;

import java.util.Set;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button btnOn = findViewById(R.id.button);
        Button btnOFF = findViewById(R.id.button2);
        Button btnDevice = findViewById(R.id.button3);
        TextView txt = findViewById(R.id.textView);

        BluetoothAdapter objAdp = BluetoothAdapter.getDefaultAdapter();
        if(objAdp == null)
        {
            Toast.makeText(this, "Bluetooth not supported!",
Toast.LENGTH_SHORT).show();
        }

        btnOn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                if (! objAdp.isEnabled())
                {
                    try {
                            Intent objIntent = new
Intent(BluetoothAdapter.ACTION_REQUEST_ENABLE);
                            startActivity(objIntent);
                    } catch (Exception e) {
                        Toast.makeText(getApplicationContext(),
e.getMessage(), Toast.LENGTH_SHORT).show();
```

```
                        throw new RuntimeException(e);
                    }

                    Toast.makeText(MainActivity.this, "Bluetooth
Enabled!", Toast.LENGTH_SHORT).show();
                }

            }
        });

        btnOFF.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                objAdp.disable();
                Toast.makeText(MainActivity.this, "Bluetooth
Disabled!", Toast.LENGTH_SHORT).show();
            }
        });

        btnDevice.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                if ( objAdp != null && objAdp.isEnabled()) {
                    Set<BluetoothDevice> pariedDevices =
objAdp.getBondedDevices();
                    if(pariedDevices.size() > 0)
                    {
                        for(BluetoothDevice device : pariedDevices)
                        {
                        txt.append("Name: " + device.getName() + "\n");
                        }
                    }
                }
            }
        });
    }
}
```

OUTPUT:


11:24
60%
Bluetooth TURN ON
Bluetooth TURN OFF
Get Devices
An app is asking to turn on Bluetooth.
Deny
Allow

11:24
60%
Bluetooth TURN ON
Bluetooth TURN OFF
Get Devices
Name: SYSKA BT4070X
Name: Carvaan mini
Name: DP-79BTR
Name: Galaxy F42 5G
Name: JACKMARTIN-CINEMA
Name: Redmi Note 8 Pro
Name: HP Fitpro

4. How do you enable or disable Wi-Fi using the WifiManager class?
5. Write code to check the current Wi-Fi connection status.

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <uses-permission android:name="android.permission.ACCESS_WIFI_STATE"/>
    <uses-permission android:name="android.permission.CHANGE_WIFI_STATE"/>
    <uses-permission
android:name="android.permission.ACCESS_NETWORK_STATE"/>
    <uses-permission android:name="android.permission.INTERNET"/>

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.AppPr20WiFiMang">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>
</manifest>
```

XML CODE:

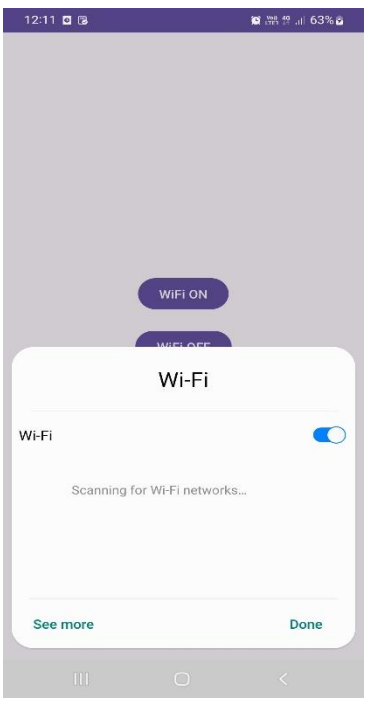
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:orientation="vertical"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="20dp"
        android:text="WiFi ON" />

    <Button
        android:id="@+id/button2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="20dp"
        android:text="WiFi OFF" />

    <Button
        android:id="@+id/button3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="WiFi State" />

</LinearLayout>
```

JAVA CODE:

```
package com.example.apppr20wifimang;

import android.content.Context;
import android.content.Intent;
import android.net.wifi.WifiManager;
import android.os.Bundle;
import android.provider.Settings;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity {
```

```
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button btnOn = findViewById(R.id.button);
        Button btnOff = findViewById(R.id.button2);
        Button btnState = findViewById(R.id.button3);

        WifiManager objWifi =
(WifiManager)getApplicationContext().getSystemService(Context.WIFI_SERV
ICE);

        btnOn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {

                //objWifi.setWifiEnabled(true);
                if ( ! objWifi.isWifiEnabled())
                {
                    Intent oIntent = new
Intent(Settings.Panel.ACTION_WIFI);
                    startActivity(oIntent);
                }
                else
                {
                    Toast.makeText(MainActivity.this, "Wifi already
Enabled", Toast.LENGTH_SHORT).show();
                }

            }
        });

        btnOff.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                //objWifi.setWifiEnabled(false);
                if (objWifi.isWifiEnabled()) {
                    Intent oIntent = new
Intent(Settings.Panel.ACTION_WIFI);
                    startActivity(oIntent);
                }
                else
                {
                    Toast.makeText(MainActivity.this, "Wifi already
Disabled", Toast.LENGTH_SHORT).show();
                }
            }
```

```
            });

            btnState.setOnClickListener(new View.OnClickListener() {
                @Override
                public void onClick(View v) {
                    int nState = objWifi.getWifiState();
                    switch (nState)
                    {
                        case WifiManager.WIFI_STATE_ENABLED:
                            Toast.makeText(MainActivity.this, "Wifi State:
Enabled", Toast.LENGTH_SHORT).show();
                        break;

                    case WifiManager.WIFI_STATE_DISABLED:
                        Toast.makeText(MainActivity.this, "Wifi State:
Disabled", Toast.LENGTH_SHORT).show();
                        break;

                    default:
                        break;
                }

            }
        });
    }
}
```

OUTPUT: